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RESEARCH ARTICLE

Mutual Consent in the Age of Smart Contracts: A Mixed-Methods Analysis of Legal Challenges

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ABSTRACT While smart contracts enhance efficiency and transparency, they raise legal and technical issues. Smart contracts do not involve face-to-face negotiation or discussion, which contributes to difficulty in confirming that both parties agreed to the terms. Moreover, while smart contracts that encode the intention of the parties show up on the blockchain as digital signatures or as preprogrammed actions, this begs the question as to precisely whether this reflects their intention and mutual consent in the first place. Furthermore, the execution of offer and acceptance in an automated manner poses a challenge to the traditional principles of contract law, as it may rely on adhesion contracts that limit the opportunities for negotiation. Moreover, the verification of legal capacity of the parties identified under a pseudonym is another challenge in a decentralized blockchain environment, especially for cross-border transactions that set varying legal standards. Through a mixed-methods approach of thematic analysis of interviews and literature review, the research responds to these challenges, across practical and theoretical domains. Proposed solutions include biometric identification, digital identity schemes, and AI-assisted consent verification. The study recommends aligning traditional legal principles with technological advancements and fostering international collaboration to create robust frameworks, ensuring fairness and enforceability in smart contracts. This study concludes that a twin-track approach—combining technological improvements with regulatory adjustments—is critical for ensuring the fairness, enforceability, and reliability of smart contracts.

INDEX TERMS Smart contracts, blockchain, mutual consent, offer and acceptance, party capacity.

I. INTRODUCTION

With the rapidly developing blockchain technology, smart contracts have revolutionized how agreements are designed and executed in the digital age [1]. Unlike traditional contracts, which are manually written and depend heavily on human involvement, smart contracts are self-executing agreements. Their terms are directly coded into the code to facilitate automatic execution as soon as pre-specified conditions are met [2]. Technology offers great advantages, such as enhanced efficiency, reduced transaction costs, and reduced reliance on intermediaries. However, it also poses complex problems, among the most pressing of which is the need to ascertain mutual consent—a cornerstone of contract law [3].

The "smart contract" term is routinely used generically within the industry but takes many forms depending on blockchain structure, execution model and transactional environment [4]. On the one hand, smart contracts could run on permissionless blockchains—such as Ethereum—permitting open, pseudonymous entry, or on permissioned ones, with access limited to known parties and often in support of regulatory or institutional supervision [5]. Second, smart contracts design range between fully on-chain where all of the logic and data are encoded on the blockchain (immutability but limited flexibility) [6], and hybrid where an on-chain component automates on-chain tasks, but it interacts

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with off-chain services (APIs, oracle services, or external databases) to model the interaction with real world events and external systems [7]. Third, the intended use case is fundamental in shaping nature and the legal implications of a smart contract. Contracts with businesses (business-to-business, or B2B) focus on integration, auditability, and operational efficiency [8]. B2C (business-to-consumer) smart contracts: These are the kind we see in Decentralized finance (DeFi), gaming or NFT platforms and here we should be even more worried about how much users understand what they are consenting to, informed consent and regulatory protection [9]. At the same time, smart contracts for governmental or public sector needs are also attracting attention, for example: land registries, public procurement, digital identity system, etc [10]. Such contracts usually operate in permissioned settings and require transparency, accountability, and adherence to public law.

To maintain conceptual clarity, this study limits our focus to smart contracts deployed in permissionless settings (including fully on-chain and hybrid), that are predominantly located in B2C and cross-border transactional settings. These factors compound legal and doctrinal uncertainties, particularly with respect to mutuality of consent, capacity verification, and enforceability. Establishing this distance in advance bolsters the analytical apparatus of this study and locates its results in the most legally and technologically disputed domains.

Mutual consent, or the ''meeting of the minds,'' is a fundamental principle of contract making, it ensures that all parties to an agreement possess a clear and common understanding of its terms. It constitutes an expression of wills, offer, acceptance, and capacity of the contracting parties, is necessary for the validity and enforceability of any agreement [11]. In conventional contracts, the consent of the parties is usually obtained by direct negotiation and communication, and thus there are chances for parties to resolve ambiguities [12]

According to Nugraheni et al. [13], The condition of MUTUAL CONSENT happens where there is a respond to the seller's offer by the buyer through a payment mechanism which represents a declaration of agreement. This situation fits with the Agreement theory. Acceptance is one of the forms of a statement of will that produces consent, as it is a statement of will of a party to approve an offer made by the other party in order to conclude an agreement or contract. The second kind of a statement of will is an offer which is a kind of a statement of will that comprises an element of initiation of an agreement. But where smart contracts are involved, the automatic and pre-programmed nature brings in questions on how actual consent is determined and if the traditional legal doctrine is in a position to manage completely the dynamics of consent within the pseudonymous and decentralized environment [3].Cutts is of the view that the nature of consent, particularly in the case of automatic agreements, can have its implications for the balance of power within contracting relationships [14]. Practical issues, such as coding errors, absence of direct negotiation, and the inability to ascertain the legal capacity of parties, also present a challenge to applying traditional legal principles to this new form of contracting [15]. Scholars maintain that smart contracts can avoid the implied processes of bargaining and understanding, and later disputes regarding whether there was actual mutual assent [16]. As Sillaber and Waltl observe, ''although a smart contract has been placed on the blockchain, this as such should not be considered a party's agreement to conclude the contract because everyone may submit every smart contract to the blockchain reflecting an obligation for whatever random wallet owner'' [17]. While Ghodoosi stressed that the problem is more acute in blockchain technology where automation, anonymity, and synchronous transactions further isolate the notion of consent [18]. The legal analysis of smart contracts, therefore, cannot be based on the notion of consent and mutual assent. They acknowledged that in smart contracts, however, the interactions that occur by means of negotiations, an exchange of promises, and mutual assent at the time of contracting are largely absent. Meanwhile, the pseudonymous nature of blockchain technology even contributes to this issue. With no apparent mechanisms to verify the identity and intentions of the parties, challenges can be raised on the authenticity of consent [19]. Alhejaili argues for regulatory rebalancing - including legal definitions of consent, capacity, protections for external data feeds, legal control of ''kill switches'' and oracles, and mechanisms for dispute resolution [20].

On the other hand, although mutual consent, or convergence of wills is a fundamental component of a contract, it is not sufficient on its own to ensure the agreement's validity, though it is adequate for establishing its existence [21]. Mutual consent, to be valid, must be a declaration of intent by each party as per requirements in law, either directly or through representation that has been authorized by the party. Therefore, the expression of wills is directly in reference to mutual consent and suggests how contracting parties express their intention of forming contractual relations [11]Traditionally, this is brought about through express communication, for example, by signing a document or agreeing orally to terms. [22]. In smart contracts, expressions of will are coded into the blockchain, more often than not, through the use of digital signatures or the undertaking of pre-agreed actions [23]. The issue is whether this coded form is sufficient for expressing the true intention of the parties [15]. Ghanmi [24] stressed as the explanation requirements for current smart contracts are four cores: justifying, clarifying, compliance, and consent, they pressed that the non-provision of these dimensions potentially raise the risks of legal and operational surprises for users. However, code-based smart contracts are not always directly akin to the types of informed and rational consent that are taken for granted in standard contract law. Therefore, embedding contract law concepts and Explainable AI (XAI) principles to address the need to explain. In addressing the doctrinal gaps cutting across smart contracts and traditional contract law, Vaishnavi and Yadav posit that the rigid automation in smart contracts erodes basic tenets of classical contract formation—such as mutual consent, intention, and capacity. Their analysis emphasizes that smart contracts—by extracting negotiation and natural language from the contracting structure—contradict the foundation of consensual agreement that underlies contract law. Of course, the authors stress that the self-executing characteristic of smart contracts is not consent unless there are mechanisms included to determine understanding and free will [25]. Coding errors or discrepancies between what is willed by the parties and what is executed by the contract can lead to disputes. Legal scholars demand attention to be paid to rendering the expression of wills in smart contracts clear and unambiguous [26]. The intention of the parties to form a legally binding contractual relationship must be determinable from the offer and its acceptance [27]. Therefore, offer and acceptance are the building blocks of mutual consent. An offer sets out the terms of the agreement, and acceptance signifies the offeree's agreement to those terms, thereby creating a binding contract agreement [11]. The way of entering into a contract is offer and its acceptance. It is noteworthy that the mutual parties' consent generally materializes by an offer submitted by one party which is accepted by another one. Substantively, the offer must include all essential elements of the contemn-plated contract outlining the intent of its author as legally binding in the event of acceptance. An incomplete acceptance of the essential terms of the offer would not form a binding contract but constitutes a mere counteroffer which would need to be accepted [27]. In traditional contracts, these elements are usually expressed through written or verbal communication, which allows for negotiation and clarification [28]. In comparison, smart contracts convert offer and acceptance into computer code, where the offer is the predetermined terms, and acceptance occurs when the pre-set conditions are met. This automation, although facilitating easier and quicker contract formation, brings about concerns of fairness and negotiation [29]. Critics have also contended that smart contracts are typically contracts of adhesion in which one party must comply with pre-arranged terms that have no negotiating space [30]. The rigid setting could lead to imbalance of bargaining power and puts into question the issue of whether smart contracts meet traditional perspectives on offer and acceptance [31].

Simultaneously, legal capacity is considered to be among the necessity requirements for the validity of every contract. It ensures the contracting parties possess mental as well as legal capacity to make sense of the terms, as well as obligations of the contract [32]. Capacity checks are simple in traditional environments in which parties appear physically and can present identification. In smart contracts, however, the pseudonymity and decentralization of blockchain make it challenging to ascertain whether parties possess the legal capacity requirements [33]. The challenge is particularly sharp in cross-border transactions, where variations in legal requirements complicate the verification exercise further [34].

Proposed solutions to manage mutual consent within the purview of smart contracts include embedding biometric identification and sophisticated digital signatures so that it is certain that all the parties involved signify their consent to the contractual conditions. Gomila and Hinarejos present a blockchain-based multi-party contract signing solution that ensures fairness and non-repudiation without the involvement of a trusted third party [35]. Sawant and Bharadi suggest the combination of multimodal biometric authentication with permissioned blockchain-based smart contracts, emphasizing the authentication of user identity first [36]. Kalamsyah et al suggest Elliptic Curve Cryptography and Elliptic Curve Digital Signature Algorithm combined with blockchain for digital contracts with better security and less complexity compared to RSA-based systems [37]. Proposing technological solutions such as user-friendly interfaces and pre-contractual disclosures to mitigate misunderstandings [38]. Similarly, O'Shields emphasized that the code would need to be produced in natural language for a court to review as part of a dispute, since it is unlikely that courts would possess the requisite expertise to review the code directly [39]. This problem could be handled prospectively by developing and maintaining an isolated version of the code translated into natural language when the smart contract goes into effect, which could be updated as changes to it are made. This should not be burdensome to developers of this technology in that they will need to provide a natural language version of the contract to the parties to obtain MUTUAL CONSENT. Nevertheless, the literature emphasizes the need for legal frameworks to bridge the gap between traditional notions of mutual consent and the automated processes of smart contracts [40]

On the topic of offer and acceptance, scholars propose incorporating customizable templates in smart contracts, in which parties may alter the terms before execution. For instance, Clack et al propose smart contract templates as a foundation for legally-enforceable contracts, connecting legal documents and standard code through operational parameters [41]. This model strives to balance automation and flexibility, resolving the inflexibility dilemma of strict smart contracts without sacrificing their benefits. Ibáñez & Simperl suggest the use of RDF documents to express terms, rights, and conditions to enable sophisticated rights checking and contractual changes [42].

On the other hand, proposed literature solutions based on capacity include integration of digital identity platforms and biometric signature in blockchain networks for verifying the capacity of parties [43]. International standards and harmonization of legal frameworks have also been advocated by some scholars to tackle such jurisdictional variances so that capacity requirements may be imposed consistently across borders [34].

Whereas most of the previous studies have significant emphases on technological innovations such as biometric authentication, digital signatures, and smart contract templates offer potential technological solutions, they cannot fully address the inherent complexities without corresponding legal reforms, However, the contribution in this paper is to propose a twin-track approach—one combining technological improvements with regulatory adjustment—to make adaptive use of smart contracts possible.

Furthermore, this study is not purely theoretical analysis limited to conceptual model analysis and literature review. Instead, it integrates theoretical analysis and practical perspectives with a view to revealing challenges and proposing solutions for the management of mutual consent in smart contracts. By employing a mixed-method approach integrating doctrinal legal research and socio-legal methodologies, the research bridges theory and practice. It draws from the findings of (27) in-depth semi-structured interviews with attorneys, judges, and academics to explore how mutual consent is being addressed nowadays within the context of smart contracts.

The research aims to enhance understanding of the legal implications, practical obstacles, and strategies for addressing these challenges, contributing valuable insights for the legal profession. By focusing on participants' responses to critical questions, the study proposes actionable recommendations to strengthen regulatory frameworks governing smart contracts. These suggestions are particularly relevant for legal practitioners and policymakers seeking to adapt regulations to the complexities of digital environments and ensure that smart contracts function effectively in this digital era.

II. METHODOLOGY

This is a mixed-methods study that relies on qualitative and normative legal research methodologies in giving a comprehensive account of the mutual consent issues in smart contracts. The qualitative component involves carrying out in-depth semi-structured interviews with legal professionals, i.e., attorneys, judges, and academics, with a perspective to gaining pragmatic understanding of the complexities of mutual consent, offer and acceptance, and capacity determination in blockchain environments. Interview participants were selected purposefully due to their experience in contract law and familiarity with the mechanics of smart contracts. Interview data were analyzed thematically for frequent issues and trends and mapped alongside broader legal and technological contexts. The normative component considers doctrinal legal analysis, statutory comparison and reviewing existing statutes to locating the findings within the context of traditional contract law principles and identifying gaps within the existing regulatory framework. With both empirical and theoretical observations and insights derived from the interview findings, not only does this research reveal the hot topics but also provides concrete suggestions for preparing legal frameworks for the automatic and decentralized aspects of smart contracts. This two-pronged approach is an assurance that theory and practice are addressed thoroughly to bridge the gap between law and technology.

A. SAMPLING STRATEGY AND QUESTION DESIGN

Purposive sampling was used in selecting the participants who possessed the expertise and experience to address the obstacles regarding the matters of smart contracts, namely party capacity and mutual consent. The approach targeted professionals based on their role, years of experience, and specialism in contract law and the governance of smart contracts [44]. By sampling veteran practitioners, the study was able to acquire in-depth data on regulatory challenges and opportunities, and its findings were more valid and applicable.

Snowball sampling was employed to expand the population of the respondents by recruiting other legal practitioners such as attorneys, judges, and scholars. Snowball sampling is a traditional technique in qualitative research, whereby referrals from the first participants are utilized in identifying other experts who belong to the same network [45]. It is a particularly useful method when traditional methods are not viable due to challenges in accessing the target group. Its flexibility enables researchers to adapt procedures to the dynamics of the community, with diverse insights and a multi-layered conceptualization of the phenomena being revealed [46]. The approach enabled the study to include extremely knowledgeable participants, which enhanced the depth and scope of its findings.

The chosen participants had been contacted through written communication or telephone call, and they were received an introduction letter explaining the study purpose, confidentiality of information, and participants' rights. Anonymity was offered through the letter, and voluntary nature was explained in the sense of not being compelled to participate, with no impact on future benefit to those participants who were not willing to participate. Follow-up questions secured participants' understanding and consent, meeting the ethical requirement. All of these processes guaranteed the study's ethical adequacy and improved the credibility of the data collection process.

The sample was snowballed from participant referrals in order to attain more comprehensive insights of mutual agreement in smart contracts. The attorneys, professors, and judges were targeted first and asked to refer other qualified professionals. This continued until the 24th interview when saturation was achieved and an additional three participants were included, making 27. Diverging perspectives were collected until no new information or perspectives emerged recognized signal of data saturation [47]. Despite practical limitations, the approach enabled an in-depth exploration of legal challenges and possible related to mutual consent and party capacity in smart contracts. Iterative recruitment generated diverging input and strong results that were representative of complexities in the environment.

Semi-structured and open-ended questions allow the respondents to provide detailed responses, which give informative data on respondents' attitudes [48] In optimally uncovering regulatory complexities, the approach allows respondents flexibility in response to articulate their views at will. The method allows a follow-up conversational dynamic, which enriches acquired data and enables a more in-depth exploration of complex issues [49]

B. INTERVIEW PROCESS

Flexibility in choosing a convenient interview location and time was provided to participants in order to support comfort and engagement. Venues included the researcher's office (6), participants' offices (16), and neutral locations like homes, libraries, or cafés (5). This gave leeway to personal choice, promoting an inclusive environment conducive to meaningful participation.

The informed consent explaining the study purpose, voluntary participation, and confidentiality assurances on the part of the participants was signed by each participant. The participants were informed that they might omit questions they did not want to answer, which provided a safe and open atmosphere. The interviews began with demographic questions, such as job descriptions, experience, gender, and professional categories, in an attempt to put responses into context and have a more informed understanding of their attitudes.

Probing questions encouraged the participants to speak extensively about difficulties and potential solutions, providing perceptive detail. The interview was concluded by asking the participants if they had any further comments or queries and thanking the researcher for their contribution. This served to enhance the depth of data obtained and promoted an atmosphere of respect and collaboration.

Audio recording of the interviews was followed by word-for-word transcription into Microsoft Word and painstaking checking of their accuracy, with a specialist double-checking the transcriptions independently. Participants were kept anonymous, for confidentiality and also to encourage candidness, through the use of pseudonyms or numbers. The rigorous process enabled the maintenance of data integrity and allowed patterns, themes, and insights to be easily established for a concrete and thorough analysis.

C. DATA ANALYSIS

Thematic Analysis, as a systematic qualitative method, was employed in this study in analyzing interview data from legal practitioners and academics [50]Click or tap here to enter text. It is a perfect method to explore complex subjects like party capacity and mutual consent in smart contracts, considering that it reports and identifies patterns and themes in a systematic manner. Its flexibility enabled an in-depth investigation of the complex issues of smart contracts, yielding compact and interesting research results [51]

Thematic Analysis enabled systematic reporting and coding of data, unveiling the salient features of the research topic without sacrificing analytical flexibility [52]. Organized phases like data collection, coding, and theme refinement were employed to ensure relevance and clarity [53]. Emerging themes were created directly from the data through inductive reasoning, offering new knowledge and highlighting repetitive patterns. This circular process deepened the understanding of the matters of mutual consent and party capacity in smart contracts, enabling a more well-rounded perspective [54]

Thematic Analysis started with familiarization of the data, from which the foundation for initial codes, theme development, and their iterative refinement to determine patterns was established. The inductive nature allowed themes to emerge naturally, finding consistent patterns and recurring insights. The findings highlighted key legal elements and proposed enhancements to smart contracts' adaptability, particularly regarding mutual consent.

Confidentiality was maintained by safe storage of the interview data in a lockable space that was accessible only to the researcher. NVivo 14 qualitative data analysis software was utilized for sorting, coding, and identifying patterns to enable systematic data visualization and analysis (QSR International, 2010). This ensured that accuracy and interpretability were added, rendering the study findings strong, consistent, and valuable in terms of generating meaningful conclusions.

Interview transcripts were imported into NVivo systematically, allotted unique tracking numbers, and linked to demographic data for efficient organization. This structured way enabled the researcher to make comparisons between groups of participants, spot patterns, and hence brought rigor to the analysis. A dedicated NVivo project enabled systematic coding, unveiling patterns, themes, and relationships. This meticulous approach generated meaningful insights and robust conclusions, enriching the study's findings and analytical depth.

Each legal expert group contributed valuable insights into the challenges of mutual consent in smart contracts. With a systematic approach and Thematic Analysis, the study revealed the legal framework and ascertained significant regulatory reforms. Through the analysis of each participant group separately and developing themes from their data, the process effectively gained diverse perspectives, revealing areas of agreement and conflict. Such a process contributed credibility to research, allowed for well-supported conclusions, made specific legal policy recommendations possible, and facilitated an informed understanding of the subject.

III. RESULTS

Through thematic analysis of interviews with attorneys, judges and academics, we have found three significant themes which (1) challenges of mutual consent in smart contracts; (2) identification of offer and acceptance; and (3) Verifying capacity of contracting parties. The subsequent analysis synthesizes the multiple perspectives within each category, identifies common patterns as well as differences between and within the categories, and engages in a critical reflection on the legal issues arising from the interview data.

A. CHALLENGES OF MUTUAL CONSENT IN SMART CONTRACTS

When it comes to the topic of mutual consent issues, junior attorneys have divided views on whether the requirement of mutual consent is a smart contract issue. (P1) had the view that it is a possible issue because there is no continuing negotiation and human interaction in smart contracts that may ensure a party complete understanding of the terms. This deficiency may result in disputes on whether or not there was legal consent, for example, caused by confusion or coding mistakes. (P2 and P3) did not, nevertheless, view mutual agreement as a significant problem. (P2) believed blockchain can ensure consent by verifying the coincidence of the two events and decency of their behavior. (P3) also assumed bilateral consent to be garnered by verification mechanisms of blockchain platforms, through which users willingly make or accept offers, presuming their intent is valid and informed. Moving on to mid-career Attorneys' views, there is more overall recognition of the requirement for mutual consent in smart contracts, although with greater focus on how problematic it is. (P4) did note that there had to be mutual consent in smart contracts, as in regular contracts, but their concern was with legal capacity. (P5) also agreed there must be agreement between the two and suggested blockchain platforms must ensure against consent deficit such as fraud or coercion. They suggested making knowledge of terms by both parties a condition precedent to execution so that disputes in this regard are obviated. (P6) indicated that free consent in traditional contracts is usually through direct communication, and its ascertainment in smart contracts, especially for cross-border applications, is laborious. They recommended that blockchain platforms adopt a mechanism for generating free consent, i.e., typing a specific code, so that parties give their consent freely and without defects. Simultaneously, senior attorneys identified the challenge in ensuring mutual agreement to smart contracts since their conditions are executed automatically. (P7) emphasized that smart contracts may be enforceable despite the fact that they don't reflect parties' actual intentions, particularly where misunderstanding or force is involved. Though such contracts are cancellable under conventional contract law principles, their automation by smart contracts renders it complex. (P8) indicated that, while smart contracts execute predetermined terms, delineating actual will as well as consent, especially complex circumstances, human intervention may be required. Data analysis and AI are, however, going to render such a process of consent easier and more straightforward. (P9) pointed out that bilateral consensus in smart contracts is harder to attain than in traditional contracts due to the pre-agreed conditions. The challenge is in ensuring that both parties comprehend and consent to the terms in whole and also addressing unforeseen circumstances or variance in interpretation leading to dispute as to whether actual consent was given.

Meanwhile, judges from the Court of First Instance (P10, P11, and P12) and the Court of Appeal (P13, P14, and P15) concurred that smart contracts automate the elements of "offer" and "acceptance." According to them, offers occurred by way of predefined conditions on the blockchain, which they accept through the satisfaction of such conditions. While this automates contract facilitation, the judges continued that this renders it problematic to ensure simultaneous agreement. Judges (P10, P11, and P12) are concerned that the rigid structure of smart contracts will not be capable of handling the subtleties of mutual consent, which can lead to miscommunication or errors in the code. Similarly, (P13 and P14) stressed that the inability to change terms after execution makes it more difficult to ensure mutual consent, whereas (P15) advocated that the blockchain platform, which would be responsible for governing these contracts, must implement mechanisms for agreements' authenticity and integrity verification. Such arguments on the consent of the parties to smart contracts were raised by the judges of the Cassation courts. (P16) raised the necessity to adhere to the provisions of general civil code, capacity of contracting parties, and absence of vices of fraud or coercion, as there has to be a means within the blockchain platform for determining free will of entry into the contract and validity. (P17) also added that, since smart contracts entail absent parties' agreements, free mutual consent will more readily be assumed, subject to defects such as fraud or coercion, which are dealt with under Civil Law. (P18) spoke of the difficulty in determining free mutual consent, particularly where there are misunderstandings or contractual terms become unfair due to unforeseen circumstances.

On the one hand, assistant Professors emphasized the urgent necessity to obtain valid consent in the context of smart contracts. Their suggestions ranged from traditional protection mechanisms, such as protection against coercion and undue influence, to new technical suggestions like the integration of electronic signatures to acknowledge digital consent. Associate Professors, by contrast, presented a more disjointed picture. Some of the concerns that were pointed out dealt with the nuances of electronic consent, emphasizing the need for clarity and specificity in the mode of expression and verification of consent in an electronic environment. Others, however, rejected special considerations in this respect, arguing that civil law resolves the issue of consent in electronic contracts adequately by not differentiating them from paper contracts. Full Professors, on their part, acknowledged the inherent complexity of smart contracts, and tended to address the interplay between their technical nature and the legal requirement of bilateral consent. A Full Professor specifically highlighted the legislative gap on mutual consent in smart contracts, i.e., the lack of general legal rules on this aspect.

All in all, the results highlight a general agreement on the role of mutual consent in smart contracts, but how strict the requirement should be and how it should be pursued remain open issues. The focus of legal attorneys and judges who would put greater demands upon safeguards in the form of judicial construction, legal modification, and statutory change, are often in favor of mechanisms that reflect similar concepts of voluntariness, informed consent, or

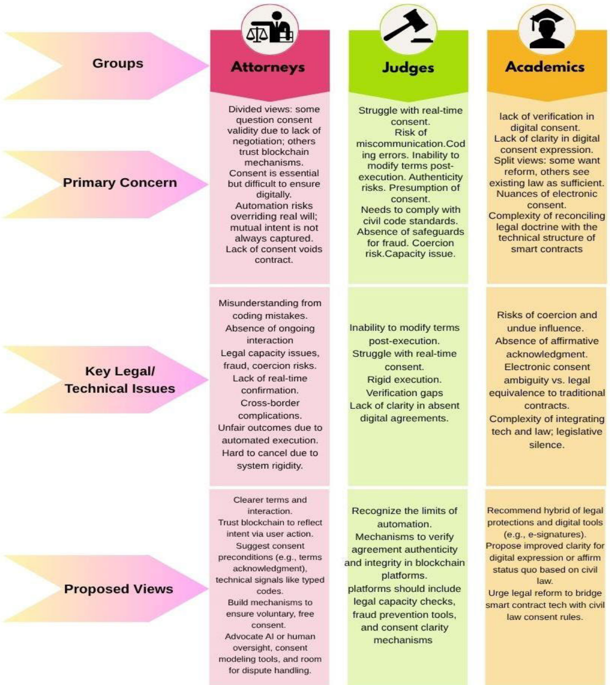


FIGURE 1. **Comparative thematic table summary of challenges in mutual consent.**

procedural fairness. Concurrently, academics are usually targeted reconciling the traditional principles of obligation with the functioning of smart contracts; underlining among the technical solutions (e.g. digital signatures and verification tools) and the regulatory lacunae that require to be addressed. This multi-layer discussion leads to the conclusion that there is a need for a hybrid legal-technical approach which allows us to maintain the ideal value of individual legal autonomy and legal safeguards when using blockchain platform while taking the advantages of blockchain capabilities to facilitate and automate the contract process.

To promote the accessibility and analytical transparency of the qualitative findings, a number of comparative thematic summaries of challenges of mutual consent are presented in the table below. By facilitating perspectives in organized and comparative retention and allowing for greater insight into the doctrinal and practical divisions that exist among professional categories, thereby enhancing the sophistication of the analysis to be deeper and more interdisciplinary.

B. IDENTIFY OFFER AND ACCEPTANCE IN SMART CONTRACTS

Several participants share the same views. Junior attorneys agreed that "offer" is the entry of terms and conditions into the blockchain by the offeror, and "acceptance" is the acceptance of the terms by the offeree and the initiation of automatic execution. All of them noted that offer and acceptance are publicly recorded on the blockchain, creating proof of agreement. Similarly, senior Attorneys described the process as automatic whereby "offer" is ascertained by pre-programmed conditions within the contract code and "acceptance" occurs when the offeree satisfies these conditions, and automatic execution is triggered. Nonetheless, (P7) referred to the use of cryptographic keys for electronic signatures, while (P8 and P9) emphasized the operation of pre-programmed conditions. Mid-career Attorneys did not in agreement. (P4) argued that the process of smart contracting to the point where offer and acceptance are incomplete or invalid because they do not entail negotiation, which may lead to adhesion contracts where the offeree will be required to stick to all of the terms as they are put forth. (P5 and P6), however, viewed the process in the classical sense, where offer and acceptance are complete when there is mutual agreement to the terms recorded on the blockchain. Therefore, when these two intentions meet and the terms of the contract, as set out in the conditional statement, are satisfied, the contract is formed between the parties.

Judiciaries of different levels of courts find the elements of "offer" and "acceptance" of smart contracts to be in accordance with traditional principles of contract but through automatic and coded mechanisms. The offer is made when the offeror codes and publishes terms on the blockchain, and acceptance is published when the offeree accepts the terms, and automatic performance follows. As precise as this is and as less uncertain it makes things, there are issues about verification of consent, potential coding errors, and absence of negotiation room, thereby rendering some smart contracts quite similar to contracts of adhesion. Judges emphasized that this necessitates careful legal consideration to surmount the challenges for the validity of consent and fairness in contractual relations.

On the other hand, academic scholars are divided in their views. Assistant Professors, for instance, focused on synthesizing traditional contract theory and technological development, citing blockchain in describing such dimensions. Associate Professors pointed out that offer and acceptance implicit in the smart contract code aligning it with both technological and legal grounds. Full Professors had varying insights, some stressed dynamic functionality of smart contracts and others doubted whether automatic processes are necessarily contracts.

In sum, the results show a strong recognition that smart contracts meet the requirements of the formality rule of offer and acceptance due to their automatic performance by operation of code. Nevertheless, such acknowledgement contains substantive concerns that are not only formal, namely on the fair, voluntary and informed nature of consent. Respondents identified risks including lack of negotiating, adhesion-like contract structures, and the risk that misinformation or misunderstanding could propagate unchecked through the inflexibility of code. These issues highlight the narrowness of a mechanistic understanding of contract formation. In this regard, the results suggest an approach for a nuanced, interpretative framework, which would not only recognize the technical soundness of blockchain contracts, but

	Attorneys	Judges	Academics
Groups			
Core Perspective on Offer & Acceptance	We have three views: 1. Offer = entering terms into blockchain; Acceptance = triggering automatic execution by offeree. 2. No real offer/acceptance—absence of negotiation = defective formation. Thus , agreement is reached when terms and intentions are coded and accepted. 3. Generally, align with classical contract logic, adjusted for automation: Offer = coded terms; Acceptance = satisfaction of coded conditions triggering execution.	Split views: 1. Smart contracts satisfy traditional definitions of offer and acceptance via automated technical means. 2. Unclear who makes the offer when code is deployed, difficulty tracing initiation. 3. Clicking or interacting with a platform may be construed as acceptance, but legal clarity is lacking.	Divergent views: 1. Generally, accept coded offer and acceptance as aligning with legal requirements. Seek to reconcile smart contract functions with classical contract doctrines. 2.Offer may be embedded in code but traditional analysis is difficult to apply. Acceptance is event-driven, raising issues about conditional or unintended consent. 3. Some affirm dynamic self-executing functionality, others question whether automation fulfills the essence of contractual agreement.
Key Legal/Technical Issues Identified	Process creates transparent record of mutual consent on-chain. Concern over adhesion contracts where one party must accept all terms without bargaining. Emphasis on electronic signature and pre-programmed rules. Consent as verifiable once blockchain conditions are satisfied. Highlight legal recognition via cryptographic tools and coded logic.	Raise issues of consent verification, coding error, lack of room for modification or negotiation. Warn that some smart contracts resemble contracts of adhesion, raising fairness concerns	Analyze blockchain as reinforcing classical elements while demanding interpretive updates. Reflect confidence in legal-technical convergence. Debate centers around whether automation alone satisfies mutual intention. Emphasize doctrinal synthesis and legal adaptation to tech realities. Highlight the dual nature (legal + technical) of smart contract operations.
Evidence of Intent	Emphasize evidentiary strength via blockchain's immutability. Need for supplementary evidence or logs to verify user intent.	Concerned about the lack of clear indicators of intent or understanding.	Caution against over-reliance on automation without verifying human intent. Suggest integrating oracles and user logs to interpret intention more accurately.

FIGURE 2. **Comparative thematic table summary on identifies offer and acceptance in smart contracts.**

also would maintain harmonization with the underlying principles of contract law, such as autonomy, fairness and mutual reliability.

To improve the clarity and accessibility of the qualitative findings, the following table presents a comparative thematic summary focused on the identification of offer and acceptance in smart contracts. By systematically organizing the perspectives of different professional groups, the table highlights key doctrinal and practical distinctions, offering a more nuanced understanding of how traditional legal concepts are interpreted in the context of blockchain-based contracting. This structured presentation strengthens the analytical depth of the study and underscores its interdisciplinary relevance.

C. VERIFYING CAPACITY OF PARTIES

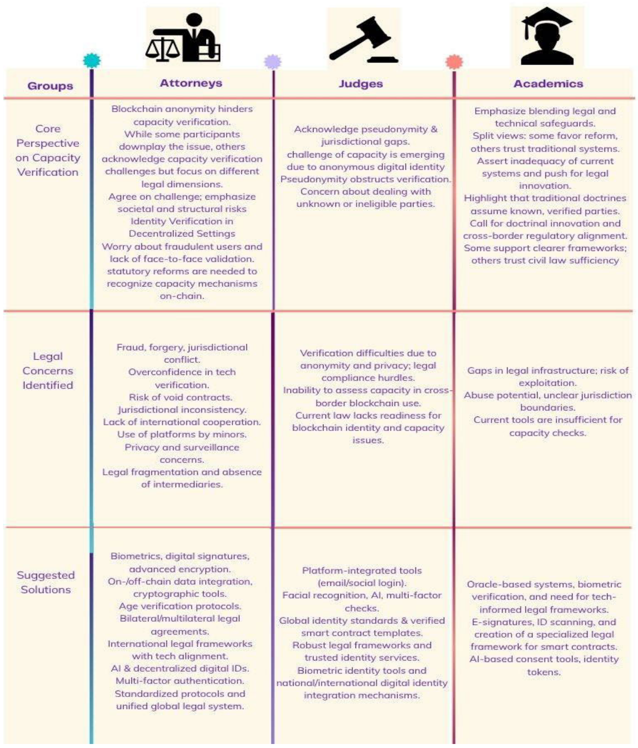
All groups of attorneys identified difficulties with ascertaining capacity on blockchain platforms since they are anonymous and decentralized. Junior attorneys (P1 and P3) both shared the view that ascertaining legal capacity on blockchain platforms is extremely difficult due to the fact that digital identities are pseudonymous and anonymous. Both of them referred to problems with fraud, forgery, and cross-border legal complications. (P2) took another view, downplaying the issues of blockchain platforms in confirming legal capacity. In contrast to (P1 and P3), who saw high hurdles, (P2) believed blockchain platforms already have robust mechanisms, such as cryptographic techniques and digital identity frameworks, to conduct verification, although they acknowledged the negligible chance of ineligible entities slipping through the cracks. While (P1 and P3) called for more robust solutions such as biometrics authentication and digital signatures, secure access controls, and yet more advanced encryption methods, legal and constitutional demands, (P2) indicated existing infrastructure as being largely adequate and suggested on-chain and off-chain data aggregation for better accuracy. Among mid-career attorneys, (P4) is most concerned with the legal implications of void contracts if legal capacity is not determined, (P5) with jurisdictional complexity, and (P6) with the need for international cooperation. Although they all agreed on the problem, their proposed solutions tackle different aspects, with (P4) demanding age verification, (P5) cross-border agreements, and (P6) international frameworks supported by technological innovation. For senior attorneys, (P7) mentioned the misuse of digital identities by kids and adolescents, (P8) mentioned privacy concerns, and (P9) mentioned the lack of intermediaries and non-homogeneous legal systems. They also suggested several solutions: (P7) called for AI and decentralized identities, (P8) requested for multi-factor authentication, and (P9) called standard protocols and homogenized legal systems on a worldwide scale.

While all judges acknowledged the difficulties posed by the pseudonymous nature of blockchain and the variations in global jurisdictional laws, their perspectives about the solutions vary. (P10) suggested developing user-friendly tools within the platform, such as email or social media profile verification, and using blockchain-based digital IDs that allow users to control and selectively disclose their information. (P11) emphasized the use of modern technologies like facial recognition and digital signatures for secure identity authentication on the blockchain, along with multi-verification procedures, i.e., mobile or email verification, supplemented by AI methods. It also demanded a legal framework for the protection of personal data and privacy.(P12) was interested in creating digital identity standards in alignment with global regulations, using blockchain-based identity solutions, biometric and multi-factor authentication, and creating standardized smart contract templates with identity verification provisions for trust building and compliance with legal frameworks. Concerns about privacy, anonymity, and compliance with different legal standards are raised by judges (P13, P14, and P15). They recommended using advanced digital authentication mechanisms and having fully established legal frameworks in place to enhance security and adherence to legislation, although (P15) believes that blockchain platforms can utilize robust identity verification mechanisms and be coupled with trusted identity verification services to render the digital identities used in smart contracts more trustworthy and reliable. In contrast, judges (P16, P17, and P18) at the Court of Cassation also refer to facing such difficulties and recommended integrating digital identity systems and biometric identity verification mechanisms for the purposes of enabling efficient responses to capacity assessment issues.

On the other hand, academics advocated robust blockchain-specific solutions, i.e., Oracle applications and biometric verification, and call for a legal framework incorporating technological security as assistant professors argued. Associate professors did share concerns regarding the potential for abuse and jurisdiction ambiguity, while several scholars argued that traditional legal systems would suffice. Full professors emphasized the need for innovation in technology, such as ID scanning and e-signatures, and they fault current structures for not delivering capacity verification in a sufficient format, emphasizing the need for a special legal regime for smart contracts.

Ultimately, the results demonstrate a universal agreement among experts of different groups that content verification of the legal capacity of the parties in smart contracts is still an open obstacle as a result of the pseudonymous, borderless, and decentralized nature of blockchain networks. Participants from all corners of the legal and academic community agreed that identity verification in traditional methods are largely ineffective in the cyber marketplace. While existing cryptographic tools were seen as ‘‘bad enough’’ of a starting point by some respondents, others pointed out how they were inadequate for the prevention of fraud, age verification, jurisdictional disparities. The spectrum of proposed solutions— from biometric-check authentication, to AI-powered verification tools, to the establishment of convergent international legal standards — underscores just how fraught, and open, the complexity of issue is. Crucially, the contrast between incremental technological improvements and the need for wholesale legal reform highlights the need for a hybrid model. This work will require the collaboration of legal scholars, technologists and policymakers, to arrive at resilient, context-dependent, means of preserving contractual integrity and protecting legal capacity in decentralized digital ecosystems.

To improve clarity and accessibility of the qualitative results, the subsequent table summarizes reflections upon verifying the ability of parties in smart contracts in a comparative thematic manner. Through its structured landscape of legal, judicial and academic perspectives, the table serves the purpose of enabling a clearer perception of the doctrinal issues and practical challenges faced with capacity validation in decentralized systems. This comparative method further serves to deepen the interdisciplinary nature of the study and to identify contrasting interpretations of legal capacity in the era of blockchain contracting.



Groups	Attorneys	Judges	Academics
Core Perspective on Capacity Verification	Blockchain anonymity hinders capacity verification. While some participants downplay the issue, others acknowledge capacity verification challenges but focus on different legal dimensions. Agree on challenge; emphasize societal and structural risks Identity Verification in Decentralized Settings Worry about fraudulent users and lack of face-to-face validation. statutory reforms are needed to recognize capacity mechanisms on-chain.	Acknowledge pseudonymity & jurisdictional gaps. challenge of capacity is emerging due to anonymous digital identity Pseudonymity obstructs verification. Concern about dealing with unknown or ineligible parties.	Emphasize blending legal and technical safeguards. Split views: some favor reform, others trust traditional systems. Assert inadequacy of current systems and push for legal innovation. Highlight that traditional doctrines assume known, verified parties. Call for doctrinal innovation and cross-border regulatory alignment. Some support clearer frameworks; others trust civil law sufficiency
Legal Concerns Identified	Fraud, forgery, jurisdictional conflict. Overconfidence in tech verification. Risk of void contracts. Jurisdictional inconsistency. Lack of international cooperation. Use of platforms by minors. Privacy and surveillance concerns. Legal fragmentation and absence of intermediaries.	Verification difficulties due to anonymity and privacy; legal compliance hurdles. Inability to assess capacity in cross-border blockchain use. Current law lacks readiness for blockchain identity and capacity issues.	Gaps in legal infrastructure; risk of exploitation. Abuse potential, unclear jurisdiction boundaries. Current tools are insufficient for capacity checks.
Suggested Solutions	Biometrics, digital signatures, advanced encryption. On-/off-chain data integration, cryptographic tools. Age verification protocols. Bilateral/multilateral legal agreements. International legal frameworks with tech alignment. AI & decentralized digital IDs. Multi-factor authentication. Standardized protocols and unified global legal system.	Platform-integrated tools (email/social login). Facial recognition, AI, multi-factor checks. Global identity standards & verified smart contract templates. Robust legal frameworks and trusted identity services. Biometric identity tools and national/international digital identity integration mechanisms.	Oracle-based systems, biometric verification, and need for tech-informed legal frameworks. E-signatures, ID scanning, and creation of a specialized legal framework for smart contracts. AI-based consent tools, identity tokens.

FIGURE 3. Comparative thematic table summary on verifying the capacity of parties in smart contracts.

IV. DISCUSSION

In discussing these challenges, it transpired that the integration of smart contracts into existing legal frameworks is faced with significant challenges, namely in connection with mutual consent, the principles of offer and acceptance, and the capacity of the parties. Legal professionals indicated the complication in these challenges and that if these foundations of contract law are to be addressed adequately from the viewpoint of emerging technologies, an inclusive approach is required. Their opinions demonstrate that the gap between traditional legal theory and the innovative nature of smart contracts needs to be filled in quest for clarity, enforceability, and fairness.

It has been noticed that legal professionals have expressed considerable concern about the adequacy of mutual consent in smart contracts. While attorneys recognized the efficiency of smart contracts in automating the elements of offer and acceptance, they also warned that this automation might bypass the nuanced requirements of genuine consent, which are fundamental to traditional contract law. Specifically, the pre-set and non-negotiable nature of smart contracts can mean that parties are not in a position to comprehend or voluntarily consent to contract terms. This is compounded by the fact that it's impossible to renegotiate terms once a smart contract is triggered, something noted by judges from various courts who pointed out the risk this poses to procuring fully informed and voluntary consent. Part of them highlighted concerns about misunderstandings and errors in the code, whereas others suggested that blockchain mechanisms function effectively to manage consent. Those legal professionals advocated pragmatic solutions, such as providing customizable templates and pre-contractual negotiations, to enhance the validity of consent in such automated systems. At the same time, academics also evolved these concerns by emphasizing the necessity to preserve the sanctity of mutual consent in the virtual environment. Assistant professors, for instance, suggested that traditional assurances against coercion must be merged with modern technological devices, such as digital signatures, so that consent would be free and legally binding. Meanwhile, the associate professors expressed a range of views—some acknowledged the challenge of giving digital consent when there is no immediate human contact, but others argued that the principles of civil law can be extended to make provision for consent in smart contracts in the same way that they apply in traditional contracts. However, full professors indicated a more systematic issue: the legislative gaps that exist in jurisdictions, where the legal code does not exhaustively address the nuances of mutual consent in electronic contracts.

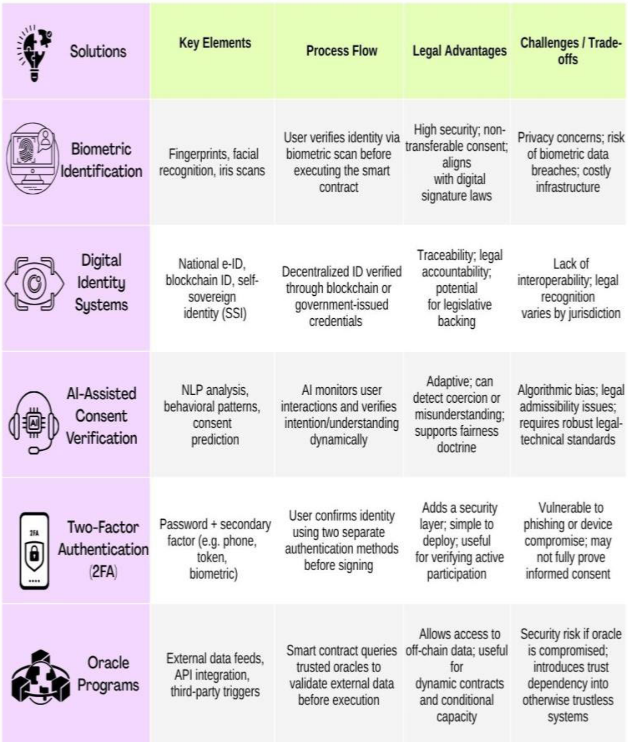
This convergence of perspectives among legal professionals underscores the need for greater assurances that consent in digital environments, such as those of smart contracts, is at least as firm and certain as in traditional contract agreements. Such assurances need to consider potential weak spots, such as misunderstandings of terms translated into computer code or gaps between automated execution and true intent of the parties. In addition, they stress the need for transparency, understandable communication, and providing means for resolving disagreements to facilitate the integrity and enforceability of digital agreements.

On the other hand, the process of offer and acceptance in smart contracts worsens the situation. Attorneys were of the view that the initial coding of a smart contract constitutes the offer, with the acceptance occurring upon execution of the coded terms at the time of execution. They caution, though, that the automated process could lead to contracts of adhesion, in which the terms are imposed without real negotiation. This concerns judges as well, who, while they acknowledged the efficiency of automated offer and acceptance, question whether the automation can ever capture the meeting of minds that constitutes the substance of contract law. Whether such automated processes may undermine the very foundations of contract formation demands a more nuanced approach to the design of smart contracts. Academic scholars, on the other hand, provided thoughtful input. Assistant professors urged a reconciliation between traditional contract theory and the need for technological applications, observing that however much smart contracts accelerate the process of offer and acceptance, this act has to function within the existing legal foundations. Associate professors largely agreed, arguing that the elements of offer and acceptance can sufficiently be addressed by the current legal framework if only it was appropriately amended. However, full professors raise serious queries as to whether the automated actions under smart contracts even constitute a contract, especially in view of the absence of immediate human interaction and bargaining.

A significant challenge that intertwines with the issues of mutual consent and offer and acceptance is verifying the capacity of the parties involved in smart contracts. Attorneys expressed serious concerns about the risks of fraud, the lack of face-to-face interaction, and the difficulties inherent in cross-border verification. Judges also highlighted the difficulties of ascertaining legal capacity in a context that is pseudonymous, where anonymity offered by the blockchain can conceal the true identity and legal status of the contracting parties. Academics added further sophistication to this discussion by highlighting the potential pitfalls and proposing solutions tailored to the specific requirements of smart contracts. To address this challenge, Attorneys, for instance, proposed undertaking high-end tech solutions such as two-factor authorization, biometrics, and global cooperation for enhanced verification of capacity of parties in blockchain platforms. Similarly, Judges also proposed inclusion of advanced digital authentication methods as well as the creation of broader legal frameworks purposed at ensuring security and sufficient check of capacity verification. They used digital identity technology and biometrics authentication methods. Meanwhile, Academics offered various views that are also reflected in the literature. Assistant professors, for instance, suggested blockchain-specific tools like Oracle programs and biometric authentication as vital to ensure the legal capacity of the contracting parties of smart contracts. Associate professors, on the other hand, pointed out the jurisdictional concerns generated by blockchain's global reach, detailing that while some existing legal frameworks can be sufficient, others require more specialized solutions. While Full professors have accused current frameworks of their shortcomings in offering capacity verification and insisted that there needs to be a more sophisticated legal approach that would be adjustable to the sophistication of smart contracts.

To better capture and compare the proposed technological solutions—from empirical data—the following figure outlines five prominent verification mechanisms: biometric identification, digital identity systems, AI-assisted consent verification, two-factor authentication, and Oracle programs. Each is assessed according to its operational process, legal benefits, and implementation challenges.

These technologies are potential instruments to enhance legal certainty in smart contracts by addressing the doctrinal necessity that consent is intentional and parties are competent. For example, biometric systems were often promoted by both attorneys and academics, because they could be used to confirm an individual's unique identity, thus ensuring that the party signing has legal personality and mental competency. An increasing number of technical/regulatory developments



Solutions	Key Elements	Process Flow	Legal Advantages	Challenges / Trade-offs
Biometric Identification	Fingerprints, facial recognition, iris scans	User verifies identity via biometric scan before executing the smart contract	High security; non-transferable consent; aligns with digital signature laws	Privacy concerns; risk of biometric data breaches; costly infrastructure
Digital Identity Systems	National e-ID, blockchain ID, self-sovereign identity (SSI)	Decentralized ID verified through blockchain or government-issued credentials	Traceability; legal accountability; potential for legislative backing	Lack of interoperability; legal recognition varies by jurisdiction
AI-Assisted Consent Verification	NLP analysis, behavioral patterns, consent prediction	AI monitors user interactions and verifies intention/understanding dynamically	Adaptive; can detect coercion or misunderstanding; supports fairness doctrine	Algorithmic bias; legal admissibility issues; requires robust legal-technical standards
Two-Factor Authentication (2FA)	Password + secondary factor (e.g. phone, token, biometric)	User confirms identity using two separate authentication methods before signing	Adds a security layer; simple to deploy; useful for verifying active participation	Vulnerable to phishing or device compromise; may not fully prove informed consent
Oracle Programs	External data feeds, API integration, third-party triggers	Smart contract queries trusted oracles to validate external data before execution	Allows access to off-chain data; useful for dynamic contracts and conditional capacity	Security risk if oracle is compromised; introduces trust dependency into otherwise trustless systems

FIGURE 4. **Comparative breakdown of technology solutions for verifying mutual consent and capacity.**

are evidence of the tool's practical feasibility and initial uptake in different jurisdictions and industries. For example, the European Union's eIDAS 2.0 regulation introduces the concept of a European Digital Identity Wallet, allowing citizens and companies to securely authenticate themselves using biometric data including facial recognition and fingerprint scans. It's a positive example of how state-backed systems are now progressing biometric ID verification in government-endorsed digital transactions [55].

Close behind were digital identity systems that merged blockchain with national or self-sovereign credentials and proved attractive to judges as something bridging the gap between technical confidence and formal legal recognition. For instance, in the private sector, decentralized identity (DID) platforms—including Microsoft's ION, Sovrin, and uPort—have operationalized blockchain-based identity management systems that eliminate the need for centralized verification authorities [56]. These systems allow individuals to manage and selectively disclose their credentials, aligning with privacy-by-design principles while supporting contractual enforceability through cryptographically verifiable digital identities.

Further, explainable artificial intelligence (XAI) and legal natural language processing (NLP) have created new opportunities for AI-facilitated consent validation These technologies are being trialed within legal tech domains to produce plain-language summaries of clauses in contracts, to track user behavior for cognitive overload or ambiguity, and to ensure that action has effectively been taken and intent expressed in the digital terms [57]. AI-assisted systems, though more contentious given concerns about bias and due process, Moore had some academics viewing them as forward-looking technologies that can simulate human-like legal judgment by analyzing behavior or the answers during the formation of a contract, or detecting subtleties like coercion or misunderstanding-matters that still escape current code-based platforms [58]. Certain solutions communicate with smart contracts via Oracle networks, allowing real time validation of the user intention following pre-defined consent rules. Additionally, integration frameworks such as the World Economic Forum's Presidio Principles and W3C's Verifiable Credentials Data Model provide architectures that can be used to securely and interoperable implement digital identity and consent tools on blockchain networks [59]

Two factor authentication (2FA) an integrated security mechanism commonly used on various digital platforms provides a practical and easily adoptable approach to bolster the identity verification. It is a process designed to authenticate a user's identity through two means (usually a one-time passwords (OTPs) and a secondary identity check through mobile or email), making it difficult for hackers to break in and pretend to be someone else [60]. Its low complexity, integrability with existing blockchain-based user-interfaces, and minimal integration cost make it an attractive choice for 2FA of permissioned smart contracts and consumer dApps [61]. Imbedded with consent procedure on executing smart contracts, this 2FA provides a mechanism of double-checking, of intentionality and protecting the ones against accidental or forced acts [62].

Valuable Oracle tools as highlighted by the assistant professors are required in any blockchain application. Bridging Information Isolation could be by enabling smart contracts to interface with off-chain data (e.g., legal databases, registry records, governmental ID verification, etc), they address the challenges of isolated information [63]. In the context of consent verification, software Oracles can supply real-time data – for example, from verified age or jurisdiction verification, external authentication Oracles – into the execution of a smart contract. For instance, a smart contract could be programmed to execute only if an Oracle confirms that the user has passed biometric verification through a trusted identity provider or has completed an informed consent module tracked by an external system. This dynamic capability is essential for enabling context-sensitive smart contracts that go beyond rigid automation and can adjust execution based on verified real-world inputs. However, they are external dependencies and add-ons which could potentially influence the trustless model in blockchain, raising concerns about verifying the authenticity as well as the origin of the data feed [64].

Despite their unique features, each solution should be seen as part of the wider landscape of legal and technical interoperability. The practicality of such tools is demonstrated by real-world deployment in fintech, healthcare and identity management systems, which in turn confirms the practicality of the recommendations provided herein. These means—no matter how avant-garde—can, as full professors stress, only work if they are effectively institutionally rooted through legal frameworks that are up to date. If there are no clear legislative to establish the probative worth of digital or behavioral information or the uniform procedures to authenticate identity, these technological developments could end up developing in a legal vacuum.

To that end, any serious reform to confirm mutual consent between and capacity of smart contracting parties should pursue a two-pronged approach: integrating these technologies into actual practice, while also updating legal rules and procedural standards so as to admit and evaluate technologically-mediated expressions of will.

V. CONCLUSION

This study highlights the most significant challenges and opportunities of mutual consent, offer and acceptance, and party capacity verification in this new arena of smart contracts. It reveals that while smart contracts have the potential to revolutionize contractual processes through automation, efficiency, and enforcement of pre-agreed terms, they also introduce significant legal and technical challenges. These complexities involve ensuring that the automated nature of smart contracts aligns with fundamental principles of contract law, such as apparent intent, informed consent, and the legal capacity of parties to enter into binding contracts. Furthermore, the study emphasizes the need to address issues related to transparency, the potential for coding errors, and mismatches between party intent and the actual execution of the contract. To fully realize the benefits of smart contracts while safeguarding fairness, enforceability, and compliance, it is essential to establish robust legal foundations and technical standards that bridge the gap between traditional contract law and emerging technological innovations.

Building on these insights, one critical issue that emerges is the assurance of mutual consent in smart contracts. The absence of direct negotiation and human interaction can result in disputes about whether parties genuinely and fully understood the terms. Judges, attorneys and academics emphasized that the rigidity of smart contracts, which prevents modifications after execution, further complicates this issue. Although blockchain activities like automated consent validation using digital signatures are somehow the solution, coercion, misunderstanding, and consent deficits are still prevalent. The legal frameworks need to embrace technical protection in terms of custom-made templates, pre-contractual disclosure, and customized mechanisms of consent validation in order to fight against such deficits.

In the meantime, the process of offer and acceptance in smart contracts also presents nuanced challenges. Although the automated execution of offer and acceptance enhances efficiency and transparency, it risks creating adhesion contracts where one party may lack the opportunity to negotiate terms. Legal experts have questioned whether automation of this sort can truly grasp the meeting of the minds necessary under traditional contract principles. The analysis indicates the necessity for an approach that balances conventional legal principles with technological innovation, such that smart contracts are brought within the parameters of the fundamental principles of contract law. This can be achieved through the incorporation of conventional offer and acceptance rules into the smart contract paradigm. Therein ensuring that encoded actions accurately mirror a clear offer and clear acceptance takes cautious coding, transparent pre-contractual disclosure, and rigorous mechanisms of determining intent. Protections must also be in place to offer remedies for probable problems, such as unintentional agreement to terms, mistakes in execution, or change of intent prior to completion. By translating the traditional rules of offer and acceptance into the digital environment, smart contracts are able to preserve the fundamentals of bilateral agreement while benefiting from the efficiency and automatization provided by blockchain technology.

Another challenge relates to the ascertainment of the legal capacity of smart contract parties. The pseudonymity and decentralized nature of blockchain environments give rise to issues of fraud, identity authentication, as well as jurisdictional clashes among legal requirements. Polyhedric technical solutions have been contemplated either by legal professionals or researchers, including biometric identification, cryptographic authentication, and digital identity programs, with a view to enhancing the validity and soundness of capacity ascertainment. Notwithstanding, this study proposes the strengthening of legal frameworks to address the conflict between the traditional demands of contract and the automated nature of smart contracts. Legal frameworks therefore must explicitly legitimize the legality of consent through automated systems. Jurisdictions must develop comprehensive legislation that addresses the nuances of mutual consent and capacity in smart contracts. This includes requiring smart contracts to file a human-readable version alongside coded terms to ensure transparency and ensure informed consent. Blockchain systems should adopt protection mechanisms such as AI-powered analysis, easy mechanisms for determining consent and capacity, and secure digital identity systems. In addition, international collaboration is required to develop harmonized legal frameworks to address the cross-border aspect of blockchain transactions and should facilitate the harmonization of regulations to ensure consistent standards for digital identity verification and capacity assessment across borders.

Ultimately, while smart contracts hold much promise in transforming contract processes through automation and efficiency, their successful adoption is predicated on a harmonized and collective effort by technologists, legal professionals, and policymakers.

For policymakers, the results point to an urgent need on the part of regulators to update the antiquated legal rules governing smart contracts. Many of the current contract doctrines are comparatively ill-equipped to parse novel contract issues such as digital consent verification, identity authentication and the enforceability trans-jurisdictional contracts. Ultimately, it is up to policymakers to ensure forward-looking legislation where the promotion of innovation and responsibility was balanced with ensuring that smart contracts will work in a system that is legally compliant and secure. Legislators are thus advised to embrace flexible forms of regulation, which consider the transient character of blockchain transactions and reflect the evolving nature of it. Those could include official endorsement of digital identity infrastructures, the promotion of cross-border legal cooperation, and new legal standards that are nimbler and technology-aware to provide sufficient confidence both in the law and in the system.

For legal professionals, the data highlight the need of reflection about how the legal concepts of mutual consent and legal capacity are adopted in decentralized and pseudonymous world. Yet legal experts should actively narrow the divide between established principles of law and the novel reality of smart contracts, by re-evaluating conceptions of mutual assent, consent, and capacity in decentralized and pseudonymous contexts. Judges and attorneys, we interviewed shared concern that traditional offer-and acceptance frameworks might not work well when applied to automated contracts and other legal instruments executed by code. Thus, practitioners are advised to familiarize themselves with smart contract architecture and the mechanics of blockchain, work with developing legal technology, and push for judicial interpretive flexibility which standardizes legal doctrine in relation to features of transactions operating in code.

For Technologists/Platform developers, the study makes an explicit contribution on how to consider legal protections when shaping system design. Developers are encouraged to create interfaces that do not obfuscate clear and informed consent, utilizing use of explainable AI, smart legal agreements, and transparent contractual terms. The addition of robust verification methods (such as biometric authentication, two-factor identification, and secure digital identity systems) can greatly enhance the legal robustness of smart contracts and also increase the confidence and trustworthiness in a platform's integrity.

By expressing these lessons learned for specific stakeholders, it is envisaged that the practicality of the study will be improved and the gap between empirical results and actual practice will be narrowed. It also highlights the need for continuous multidisciplinary cooperation to challenge the rapid developments in law and technology that smart contracts represent.

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